

Project Report: From College Boards to Dashboards

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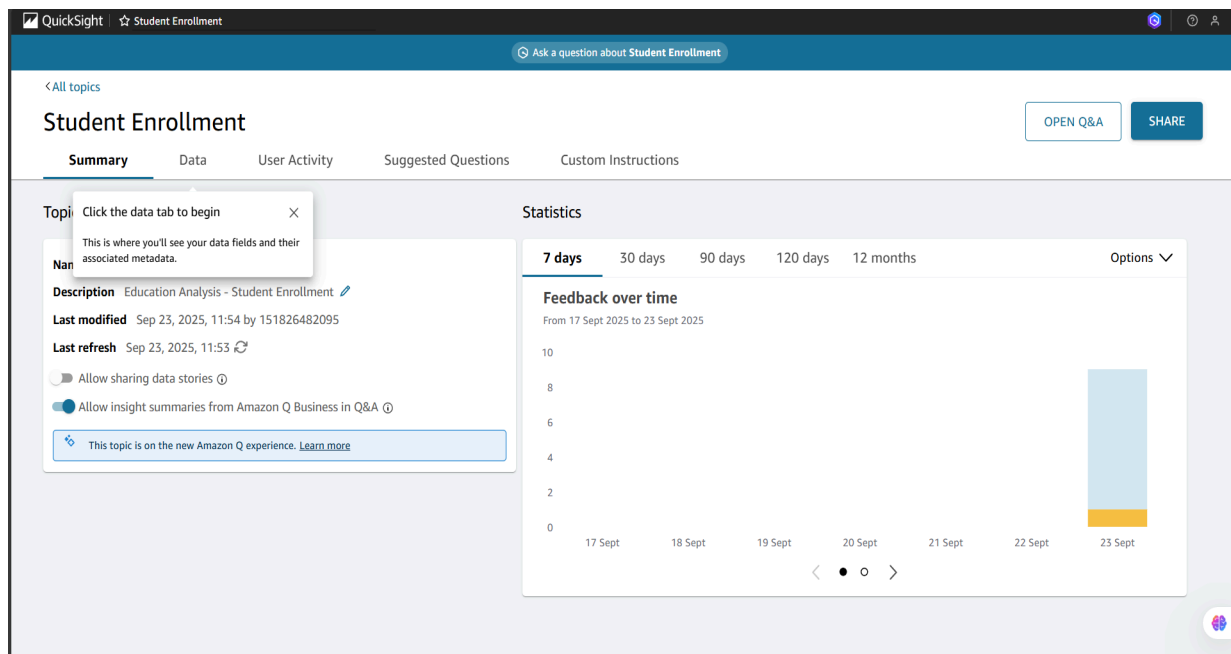
Tool Used: Amazon QuickSight **Project Title:** Providing Data Insights to a Community College

Project Overview

This project simulates a real-world business intelligence workflow using Amazon QuickSight and Amazon Q to support a fictional community college in understanding student enrollment patterns. The goal was to create, configure, and analyze a dataset, build interactive visuals, and develop a data-driven story that could inform strategic decisions for improving student satisfaction without increasing costs.

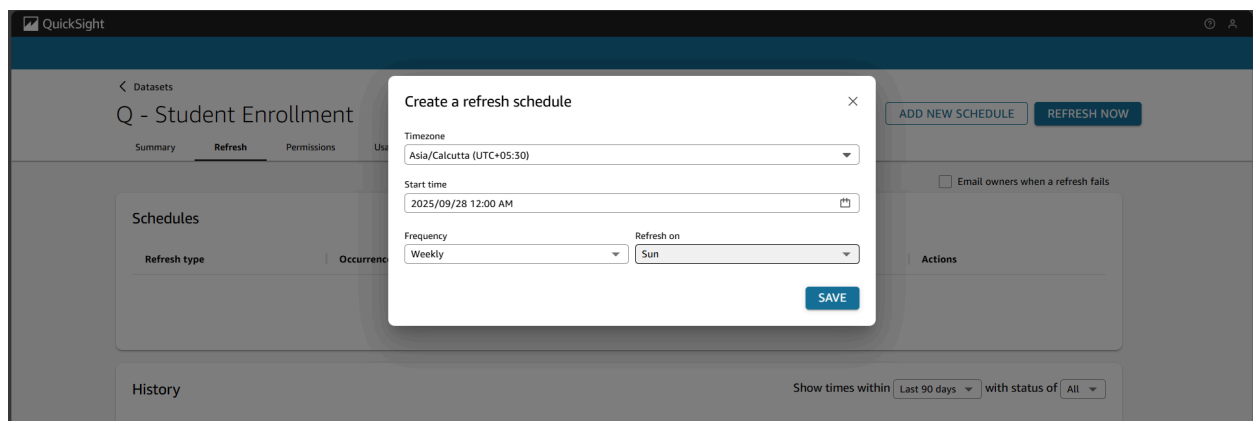
Sample Topic Setup

The first step was to create a sample topic titled *Student Enrollment*. This process took several minutes to initialize. Once completed, the topic served as a foundation for exploring student data using Amazon Q.



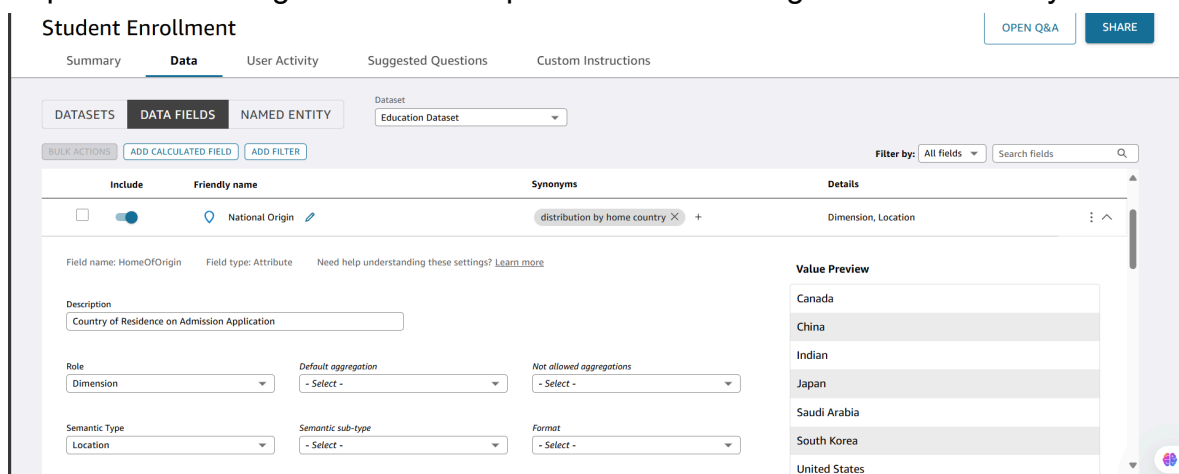
Dataset Refresh Configuration

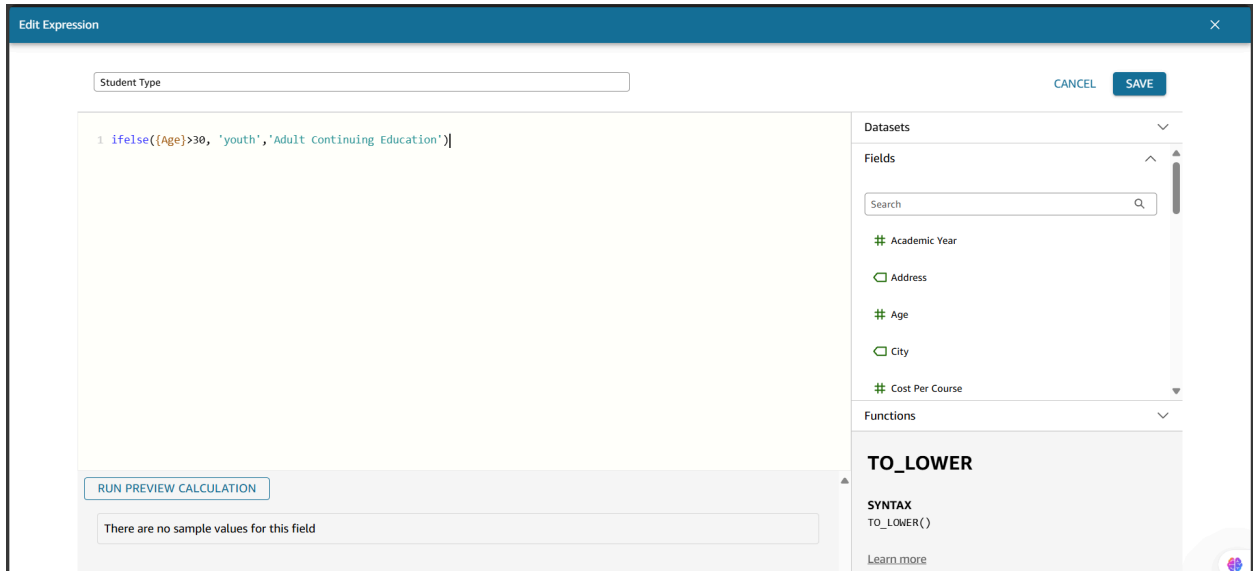
The dataset *Q - Student Enrollment* was configured with a refresh schedule to ensure data updates occurred regularly. The timezone was set to Asia/Kolkata, with the refresh scheduled to begin at 12:00 AM on the upcoming Sunday and repeat weekly. This configuration ensures that the dataset remains current for ongoing analysis.



Dataset Enhancements

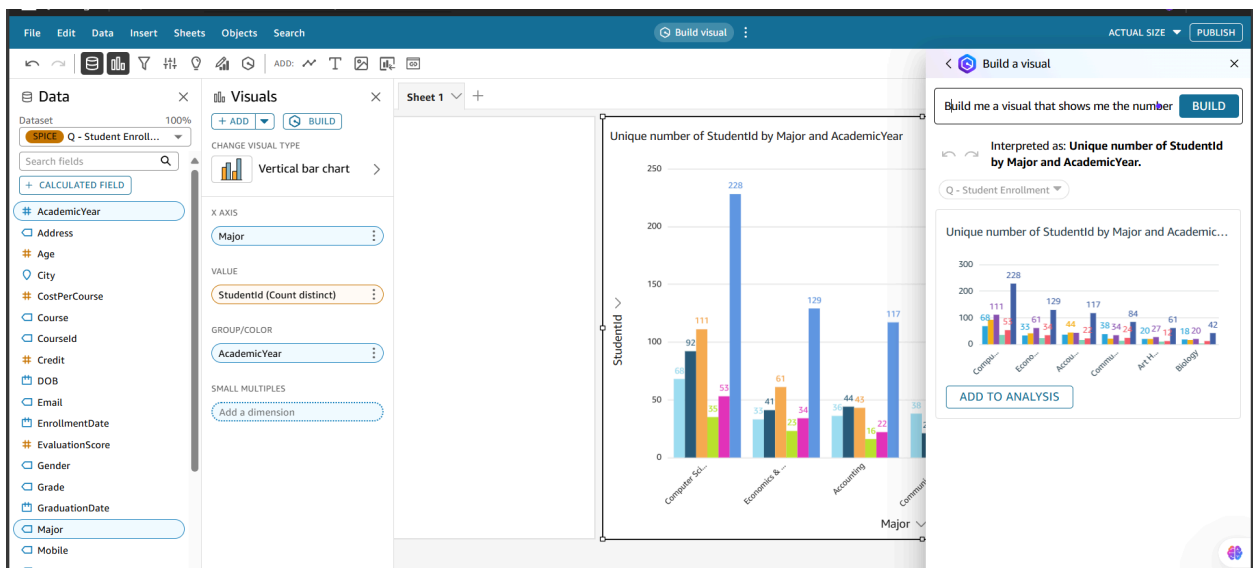
To improve clarity, the field *HomeOfOrigin* was renamed to *NationalOrigin*, with the description updated to “Country of Residence on admission application.” Additionally, a calculated field named *Student Type* was created using the formula `ifelse({Age} < 30, "Youth", "Adult Continuing Education")`. These enhancements have helped me to categorize students present in the college more effectively for analysis.

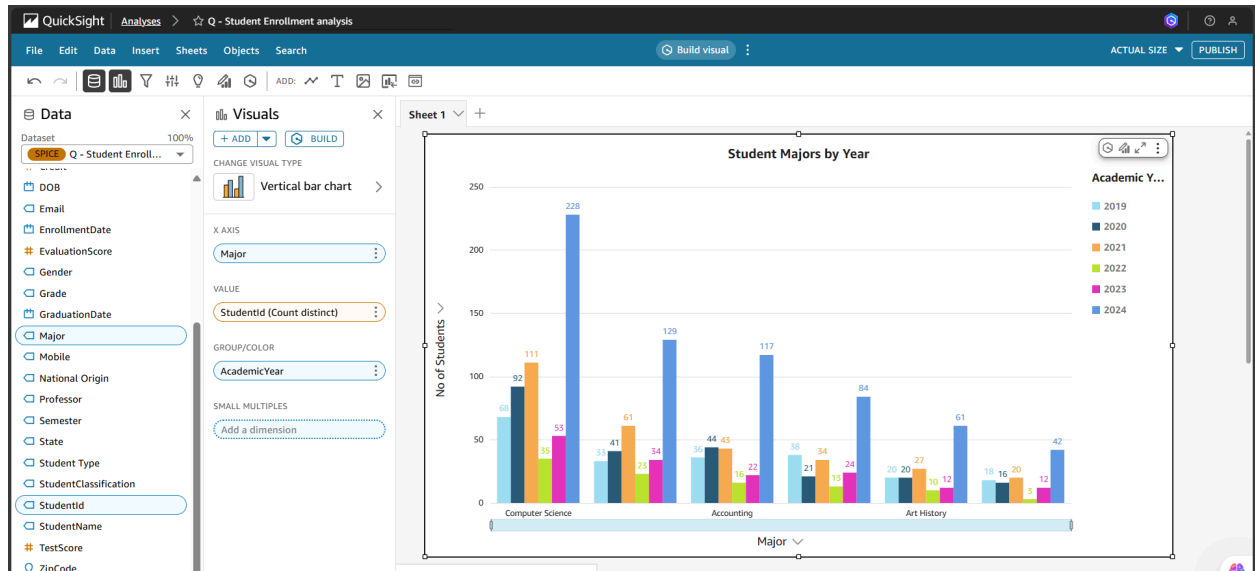




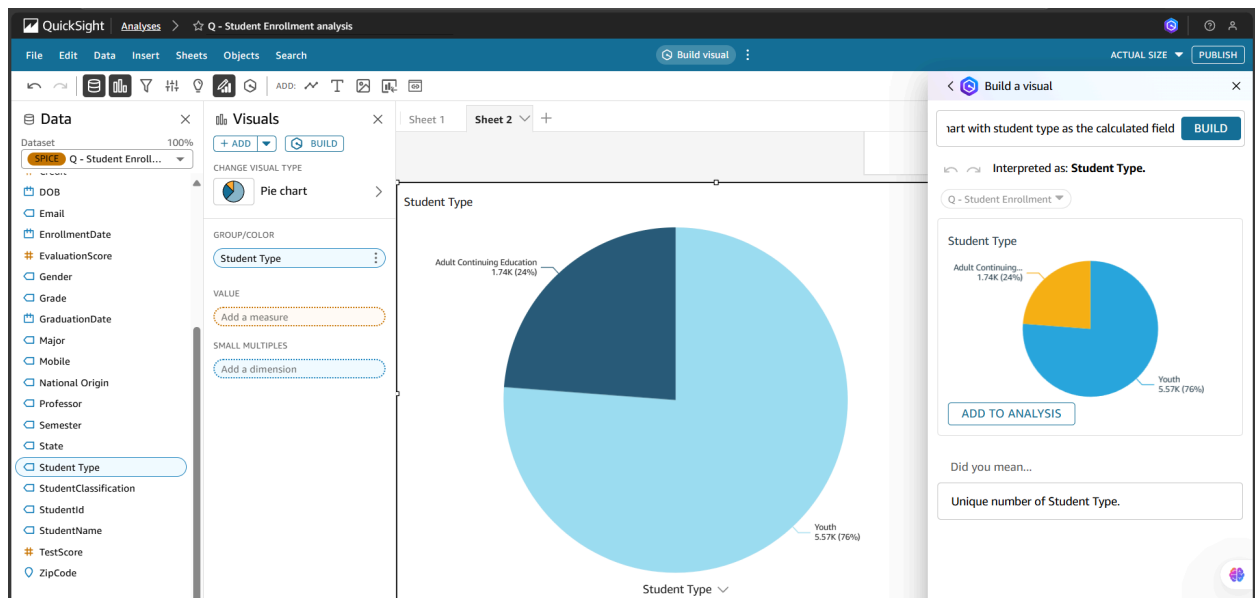
Visual Analysis Using Amazon Q

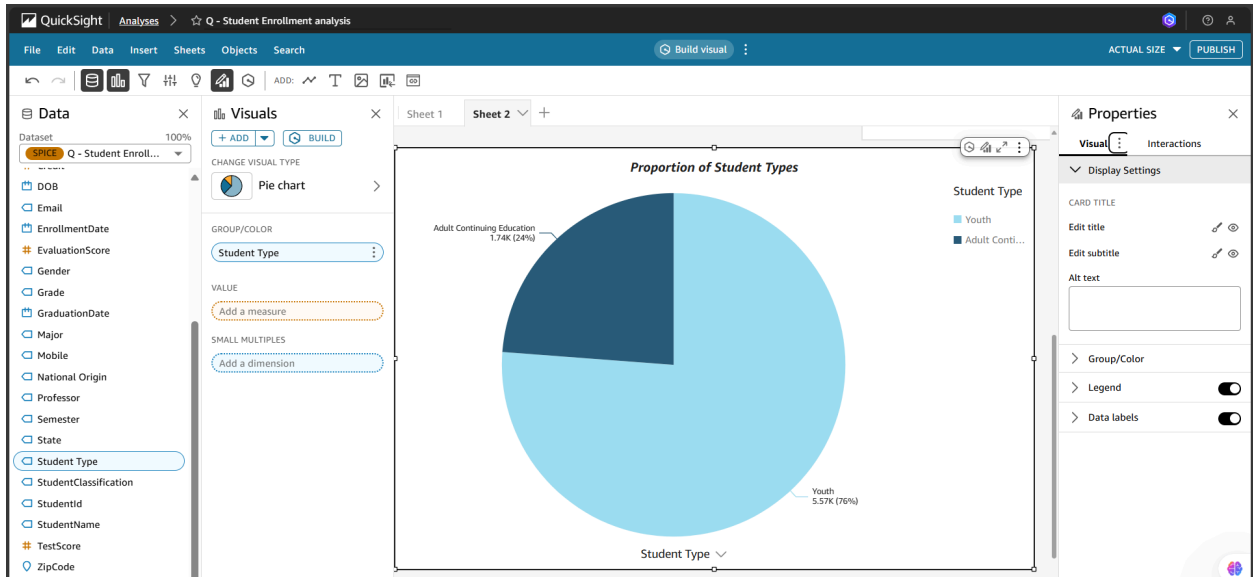
An analysis was created using the *Q - Student Enrollment* dataset. The first visual was generated using the prompt “Number of students in each major, grouped by academic year.” The chart was edited to improve readability: the title was changed to *Student Majors by Year*, the layout was resized, and the formatting of the *AcademicYear* field was corrected to remove commas. The chart type was also changed to a vertical bar chart for better visual impact.



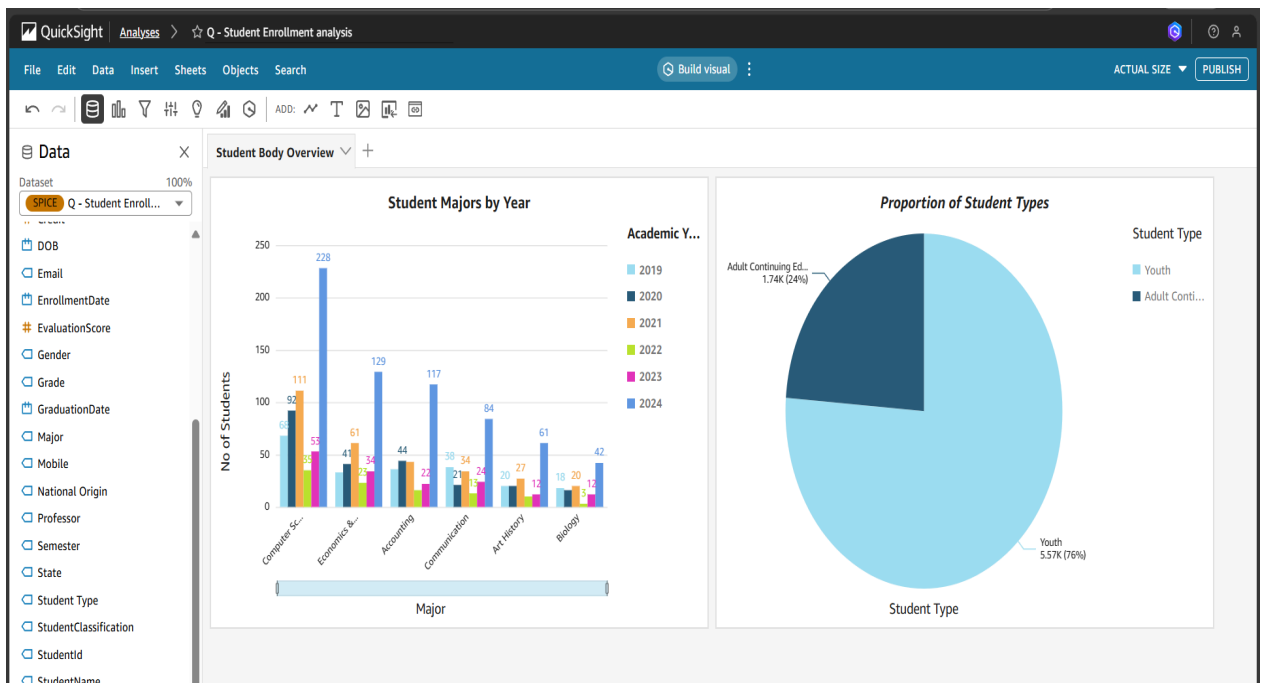


A second visual was added using the prompt “Pie chart of Student Types.” This chart was titled *Proportion of Student Types* and resized for clarity. The sheet containing both visuals was renamed to *Student Body Overview*.





The sheet containing both visuals was renamed to *Student Body Overview*.



Q Topic Configuration

The sample topic was deleted, and a new topic named *Regional Community College Student Data* was created. This topic included fields such as Student Type, NationalOrigin, Gender, Course, Grade, TestScore, and more.

The screenshot shows the 'Regional Community College Student Data' topic configuration page in Amazon QuickSight. The page has a top navigation bar with the QuickSight logo, the topic name, and a search bar. Below the navigation bar, there are tabs for 'Summary', 'Data', 'User Activity', 'Suggested Questions', and 'Custom Instructions'. The 'Summary' tab is active, showing a 'Name' field with the value 'Regional Community College Student Data', a 'Description' field with the value 'Q&A environment for understanding student enrollment data', and 'Last modified' and 'Last refresh' dates. There is a 'START REVIEW' button and a 'Review your Topic settings to improve performance!' message. The 'Topic details' section shows the 'Name' field with a pencil icon, the 'Description' field with a pencil icon, and the 'Last modified' and 'Last refresh' dates. The 'Statistics' section shows a bar chart and a pie chart.

Two named entities were defined: *Course Details*, and *Professor Evaluation*, each with ranked fields to support natural language queries.

The screenshot shows the 'Regional Community College Student Data' topic configuration page in Amazon QuickSight, specifically the 'Named entity' section. The 'Named entity' section is titled 'Course Details' and has a description 'Information regarding courses'. It shows a 'FIELD RANKING' list with six items: #1 Course, #2 Professor, #3 CostPerCourse, #4 AcademicYear, #5 Semester, and #6 CourseId. The 'Data' section on the left shows a list of fields: AcademicYear, Address, Age, City, CostPerCourse, Course, CourseId, Credit, DOB, Email, EnrollmentDate, EvaluationScore, Gender, Grade, GraduationDate, Major, Mobile, National Origin, Professor, and Semester. The 'Course Details' table on the right shows a sample view of the named entity data with columns: Course, Professor, CostPerCo..., AcademicY..., Semester, and CourseId. The table contains 15 rows of data. A note at the bottom states: 'This is a sample view of the named entity data. Complete data will be available to view in Q and used to answer questions.'

Course	Professor	CostPerCo...	AcademicY...	Semester	CourseId
Accounting	Tim	1935	2019	Fall	2019 Fall Accou...
Big Data	Jason	2496	2019	Fall	2019 Fall Big Data
Commercial Law	Sue	2540	2019	Fall	2019 Fall Com...
Communication	Antony	2536	2019	Fall	2019 Fall Com...
Communication	Antony	2536	2024	Fall	2019 Fall Com...
Counseling Psyc...	Jimmy	1812	2019	Fall	2019 Fall Couns...
Data Visualization	Kim	1244	2019	Fall	2019 Fall Data ...
Environmental ...	Jill	2856	2019	Fall	2019 Fall Enviro...
Financial Modell...	Cindy	1298	2019	Fall	2019 Fall Finan...
Financial Modell...	Cindy	1298	2024	Fall	2019 Fall Finan...
General Biology	Peter	2812	2019	Fall	2019 Fall Gener...
Investment	Tim	2404	2019	Fall	2019 Fall Invest...
Modern History	Tony	2106	2019	Fall	2019 Fall Moder...
Python2	Kim	1920	2019	Fall	2019 Fall Python2

QuickSight Regional Community College Student Data

Learn more about named entities CLOSE OPEN Q&A SAVE

Data

Dataset: Q Student Enrollment

Search fields

- Credit
- DOB
- Email
- EnrollmentDate
- EvaluationScore
- Gender
- Grade
- GraduationDate
- Major
- Mobile
- National Origin
- Professor
- Semester
- State
- Student Type
- StudentClassification
- StudentId
- StudentName
- TestScore
- ZipCode

Named entity

Entity name: Professor Evaluation

Description: Enter a brief description

Synonyms: Add alternate names for named entity

FIELD RANKING (6)

- #1 Professor
- #2 Course
- #3 Semester
- #4 AcademicYear
- #5 StudentName
- #6 EvaluationScore

Professor Evaluation

Sync table view with field ranking

Professor	Course	Semester	AcademicY...	StudentNa...	Evaluation...
Jason	Big Data	Fall	2022	Milton Treamayne	65.01
Tina	Big Data	Fall	2021	Amerigo Gegg	65.0111
Peter	General Biology	Fall	2021	Fitzgerald Borrel	65.0376
Tina	Big Data	Fall	2021	Giacobo Appley...	65.0478
Jimmy	Counseling Psyc...	Fall	2021	Amerigo Gegg	65.0495
Jimmy	Counseling Psyc...	Fall	2021	Jordain Ivashkin	65.0697
Peter	General Biology	Fall	2023	Lorenzo Potts	65.0709
Jimmy	Counseling Psyc...	Fall	2019	Abie Rentoil	65.1111
Peter	General Biology	Fall	2022	Manfred Garfoot	65.1176
Jason	Big Data	Fall	2020	Ardyth Shelsher	65.1523
Peter	General Biology	Spring	2020	Audrie Probyn	65.1801
Jason	Big Data	Fall	2022	Torrie Fossey	65.2044
Tina	Big Data	Fall	2021	Darla Forcer	65.2187
Jimmy	Counseling Psyc...	Fall	2019	Kelcy Sone	65.2257

This is a sample view of the named entity data. Complete data will be available to view in Q and used to answer questions.

Verified Q&A Setup

Using the Open Q&A feature, prompts were issued to determine which instructors received the best average evaluations and which courses were most expensive. Answers were verified and added to the list of trusted responses.

QuickSight Regional Community College Student Data

Regional Community College Student Data PINBOARD

SHARE FEEDBACK

which instructor got the best average evaluationscore ASK

Verified Interpreted as: Average of EvaluationScore by Professor

The average EvaluationScore across all professors is 75.69. There are 17 unique professors in the dataset. The top-performing professor is Jill, who achieved the highest average EvaluationScore of 78.63. Additionally, the total school year figure is 14,767,070.

REVIEW FOR ACCURACY

Average of EvaluationScore by Professor

Professor	Average of EvaluationScore
Jill	78.63
Andrew	78.47
Antony	77.91
Tim	77.87
Kim	77.73
Tony	76.54
Jimmy	76.48
Lee	75.87
Jason	75.63
Fiona	75.18

Average of EvaluationScore 75.69

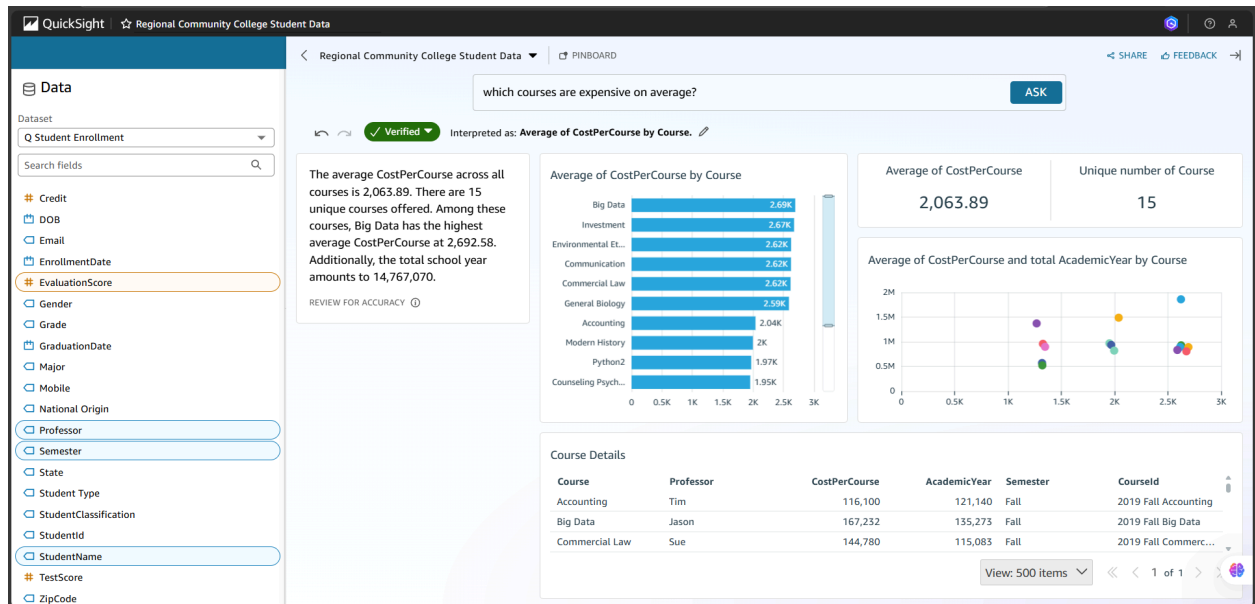
Unique number of Professor 17

Average of EvaluationScore and total AcademicYear by Professor

Professor Evaluation

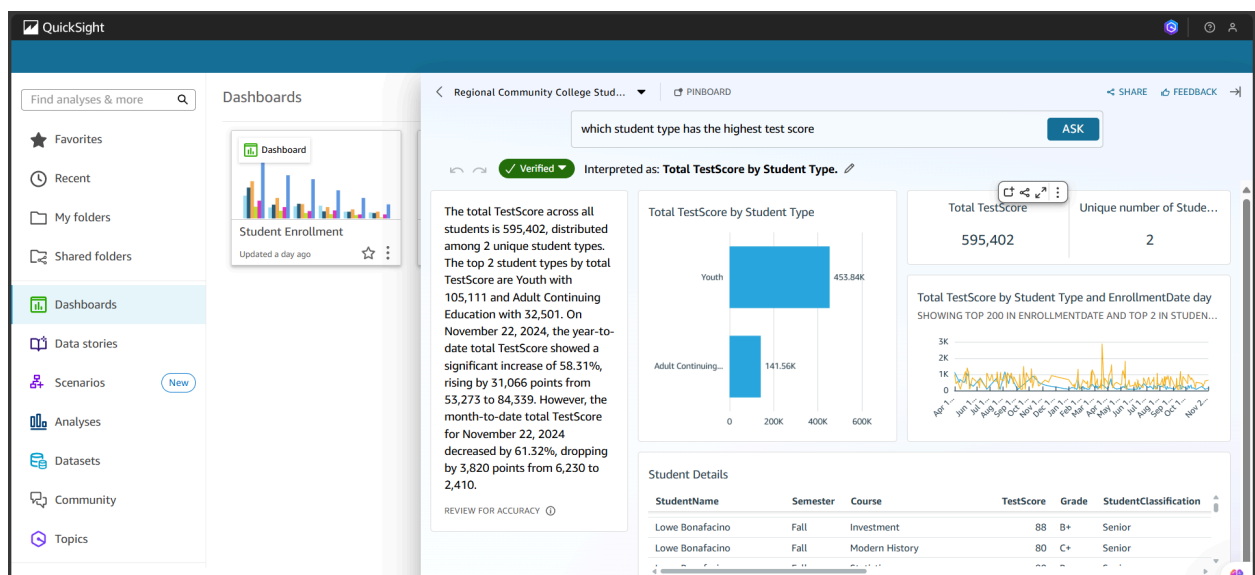
Professor	Course	Semester	AcademicYear	StudentName	EvaluationScore
Cindy	Financial Modeling	Spring	4,045	Aaren Andryushin	134.34
Jack	US History	Fall	2,021	Aaren Andryushin	68.33
Jimmy	Counseling Psychol...	Fall	2,021	Aaren Andryushin	70.97

View: 500 items 1 of 13



Dashboard Creation

The analysis was published as a dashboard titled *Student Enrollment Dashboard*, with Q&A enabled. A test prompt—"Which Student Type has higher test scores?"—was used to validate the Q&A functionality.



Scenario Development

A scenario named *Improving Student Satisfaction Without Increasing Costs* was created using visuals from the dashboard. The scenario began with the question “How do we improve professor evaluations, while avoiding an increase in cost per course?” Follow-up questions were added to explore contributing factors and refine the analysis.

The screenshot shows the QuickSight Scenarios interface. On the left is a navigation pane with options like Favorites, Recent, My folders, Shared folders, Dashboards, Data stories, Scenarios (selected), Analyses, Datasets, Community, and Topics. The main area is titled 'Scenarios' and features a 'CREATE SCENARIO' button. Below this is a card for the scenario 'Improving Student Satisfaction Without Increasing Costs', which includes a video player and a table of scenarios.

Name	Owner	Last Modified
Improving Student Satisfaction Without Increasing Costs	Me	In a few seconds

Rows per page: 100 1-1 of 1

The screenshot shows the 'Add visuals' dialog in QuickSight. It displays a list of available visuals on the left and a grid of selected visuals on the right. The selected visuals include 'Student Body Overview', 'Student Majors by Year', 'Proportion of Student Types', 'Average of Evaluation Score by Professor', and 'Average of Evaluation Score by Course'.

Find your dashboards

Student Body Overview

Student Majors by Year

Proportion of Student Types

Average of Evaluation Score by Professor

Average of Evaluation Score by Course

6 Visuals selected (Maximum 20)

CANCEL ADD

QuickSight

Scenarios

Improving Student Satisfaction Without Increasing ...

FileDataView

SHARE

Data

FIND DATA

UPLOAD FILE

Student Enrollment Dashboard

Student Enrollment Dashboard_Q - Student ...

How do we improve professor evaluations, while avoiding an increase in cost per course?

Step 1: Analyze the student enrollment dataset to understand basic statistics and data structure

Major	StudentId	AcademicYear	Student Type
Economics & Finance	24cb45a1ade144d2873737db4e16ae...	2021	Youth
Economics & Finance	fe2ca082525445a4887005b34e7fb9c5	2023	Adult Continuing
Economics & Finance	eaedbf1fe6348938711537fb3eeff3	2019	Adult Continuing
Accounting	ca1e237ecda349e289dcd5fe8318c5bd	2021	Youth
Computer Science	bc78e268e7b54c039237e635707b4...	2023	Adult Continuing
Economics & Finance	5598822dddb40038a4a8811ed539...	2021	Adult Continuing
Computer Science	ad96b22846924c429911a4424bb82...	2019	Youth
Computer Science	4d8899f78ff646778502b5e974b138d0	2021	Adult Continuing
Computer Science	0d834d6c377a8d97d99ad56fda5crr6d	2019	Adult Continuing

The dataset contains **7,306 student enrollment records** with evaluation scores ranging from **65.0 to 98.7** (average: 75.7) and course costs ranging from **\$1,206 to \$2,864** (average: \$2,064).

Step 2: Analyze professor evaluation scores and costs by aggregating student enrollment data.

QuickSight

Scenarios

Improving Student Satisfaction Without Increasing ...

FileDataView

SHARE

Data

FIND DATA

UPLOAD FILE

Student Enrollment Dashboard

Student Enrollment Dashboard_Q - Student ...

Step 2: Analyze professor evaluation scores and costs by aggregating student enrollment data.

Professor	Average_Score	Min_Score	Max_Score	Number_of_Courses
Wang	70.75	67.03	97.32	325
Jack	73.54	68	96.91	439
Cindy	73.59	67.01	96.08	1038
Jane	73.75	69.07	92.39	103
Tina	74.24	65.01	96.52	137
Peter	74.3	65.04	96.99	413
Sue	74.97	68.08	96.67	443
Fiona	75.18	67.08	97.32	247
Jason	75.63	65.01	96.81	305

Professor	Average_Score	Min_Score	Max_Score	Number_of_Courses	Average_Cost
Wang	70.75	67.03	97.32	325	1277.32
Jack	73.54	68	96.91	439	1332.77
Cindy	73.59	67.01	96.08	1038	1737.37
Jane	73.75	69.07	92.39	103	1271.26

QuickSight

Scenarios

Improving Student Satisfaction Without Increasing ...

FileDataView

SHARE

Data

FIND DATA

UPLOAD FILE

Student Enrollment Dashboard

Student Enrollment Dashboard_Q - Student ...

The analysis reveals an interesting pattern: professors with **lower evaluation scores** (around 70-74) often have **lower average costs** (\$1,200-\$1,700), while some professors with higher scores have significantly higher costs (\$2,600+).

Step 3: Identify professors who achieve above-average evaluation scores while maintaining below-average costs per course.

Professor	Average_Score	Min_Score	Max_Score	Number_of_Courses	Average_Cost
Jill	78.63	67.02	94.74	460	2624.03
Andrew	78.47	66.19	95.41	178	2695.28
Antony	77.91	66.01	98.21	920	2622.62
Tim	77.87	69	98.74	352	2388.13
Kim	77.73	67.27	97.12	315	1704.04
Tony	76.54	68.03	96.81	407	1996.24
Jimmy	76.48	65.05	97.67	480	1951.85
Lee	75.87	67.01	96.47	744	1602.58
Jason	75.63	65.01	96.81	305	2604.14

Average_score and Average_cost by Professor

3K

2.5K

Wang

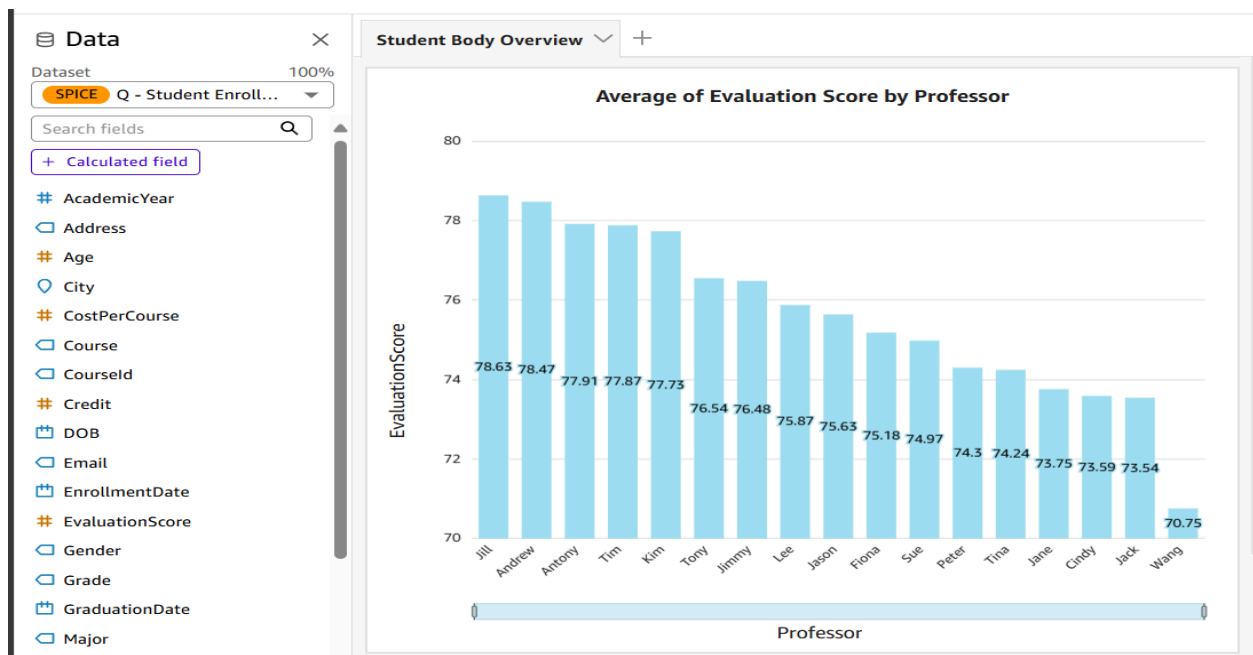
Jack

Cindy

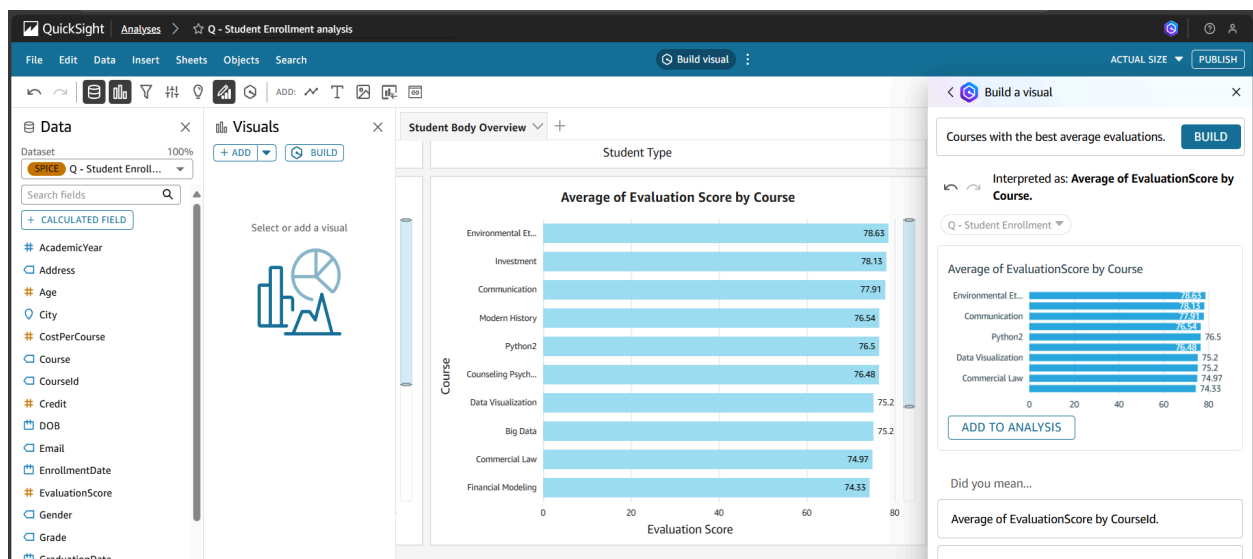
Extended Analysis

The analysis was extended with four new visuals: professors with the best average evaluations, courses with the best average evaluations, professors with the highest average course costs, and courses with the highest average course costs. These visuals were published to replace the existing dashboard and provide deeper insights.

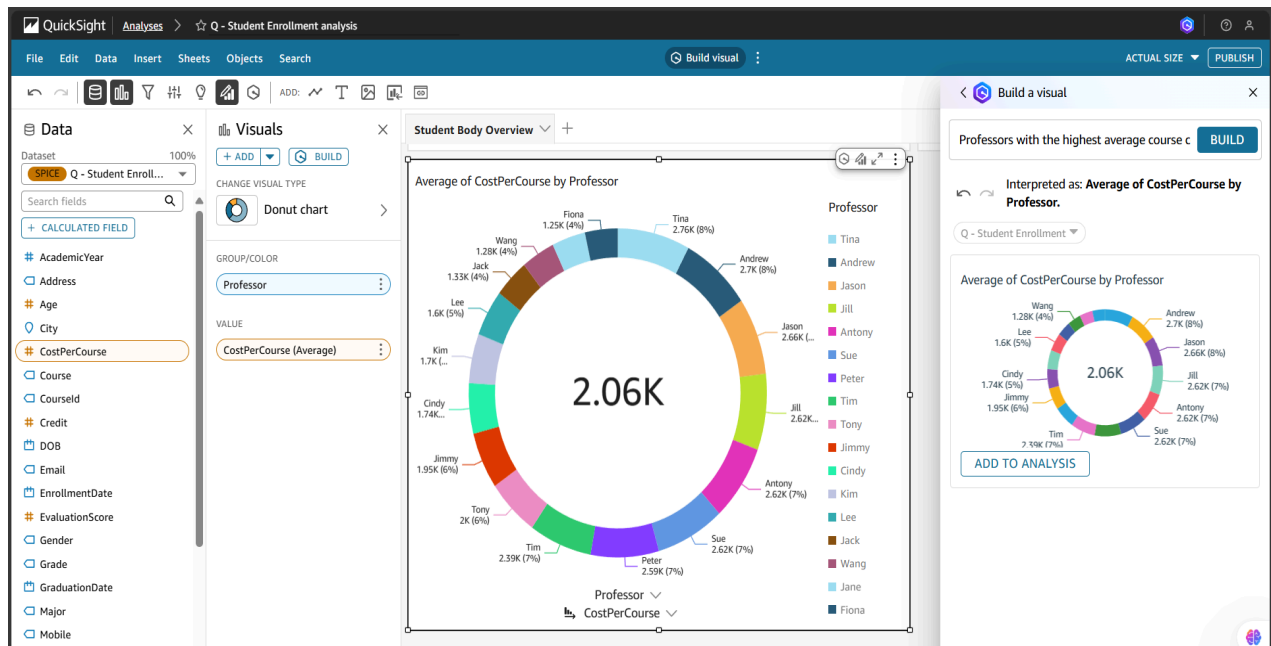
1. Professors with the best average evaluations



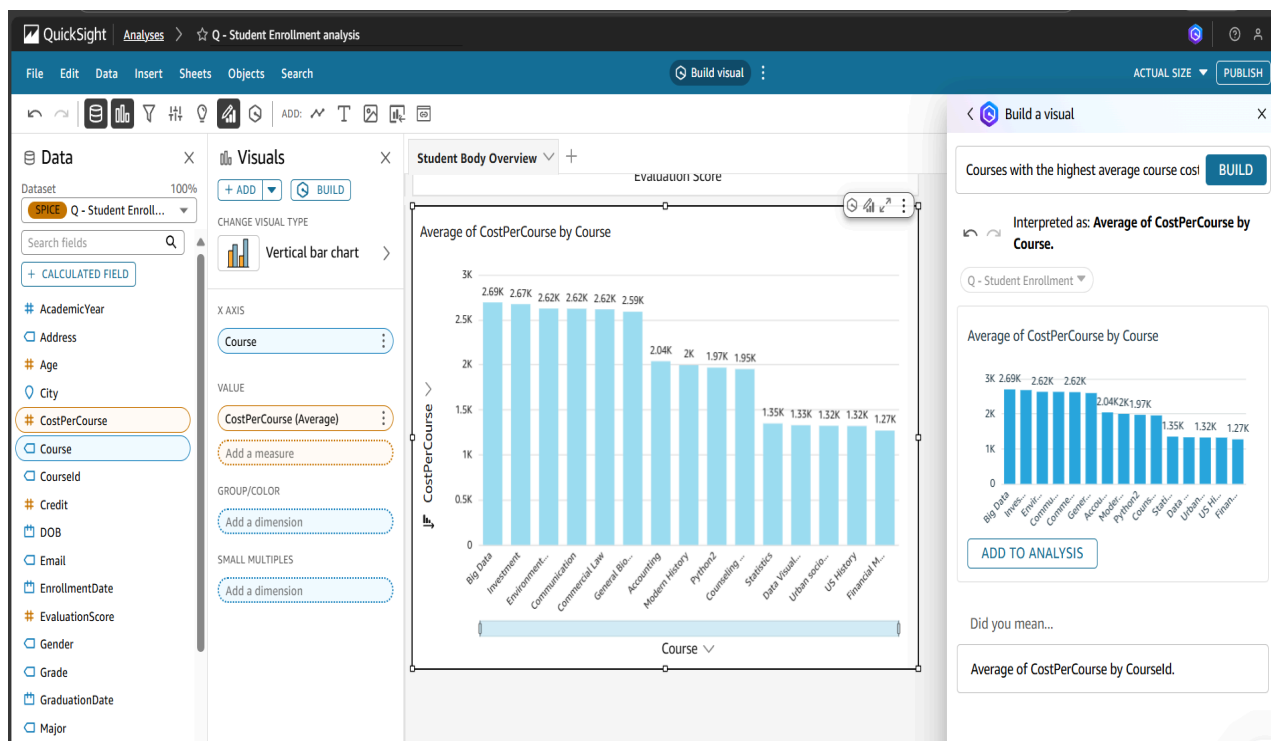
2. Courses with the best average evaluations



3. Professors with the highest average course costs

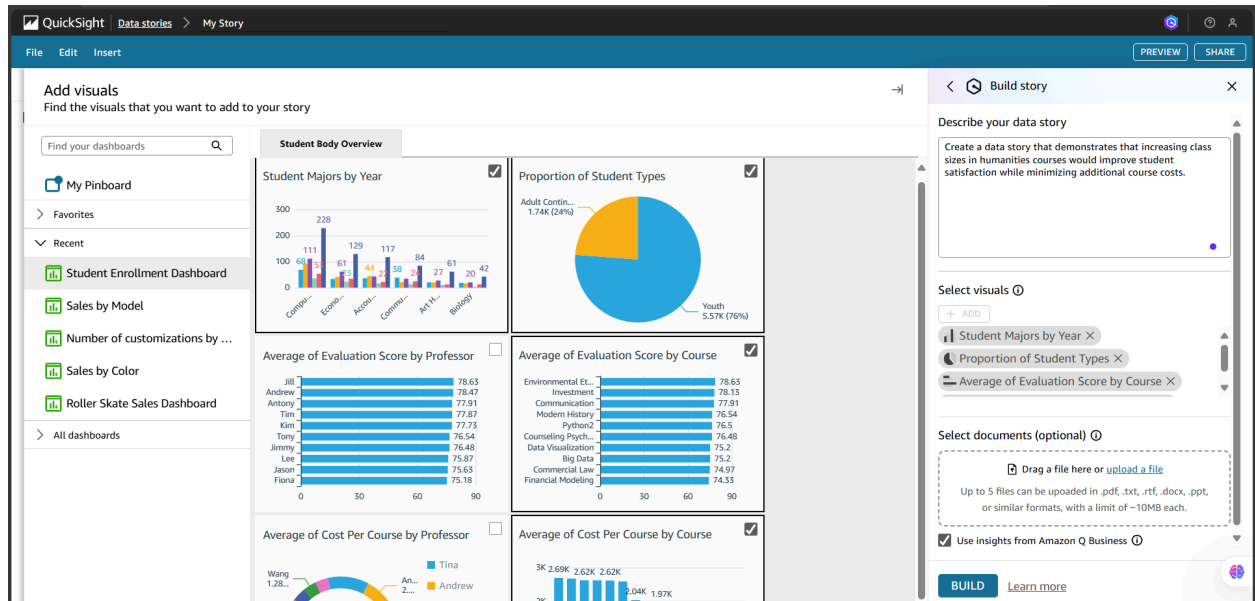


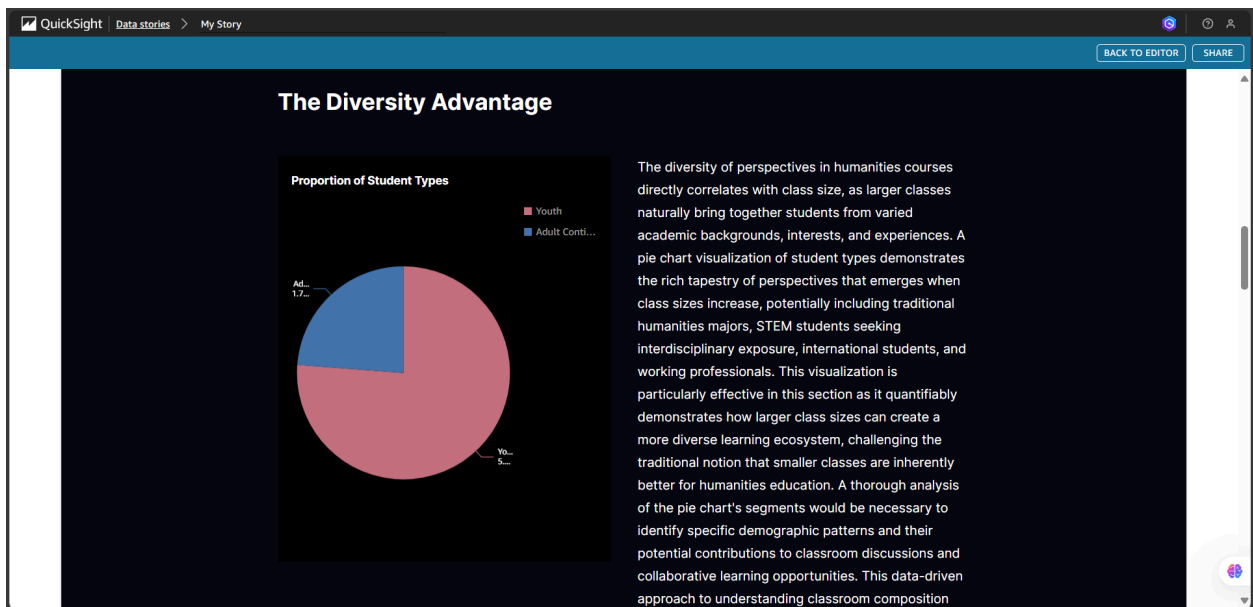
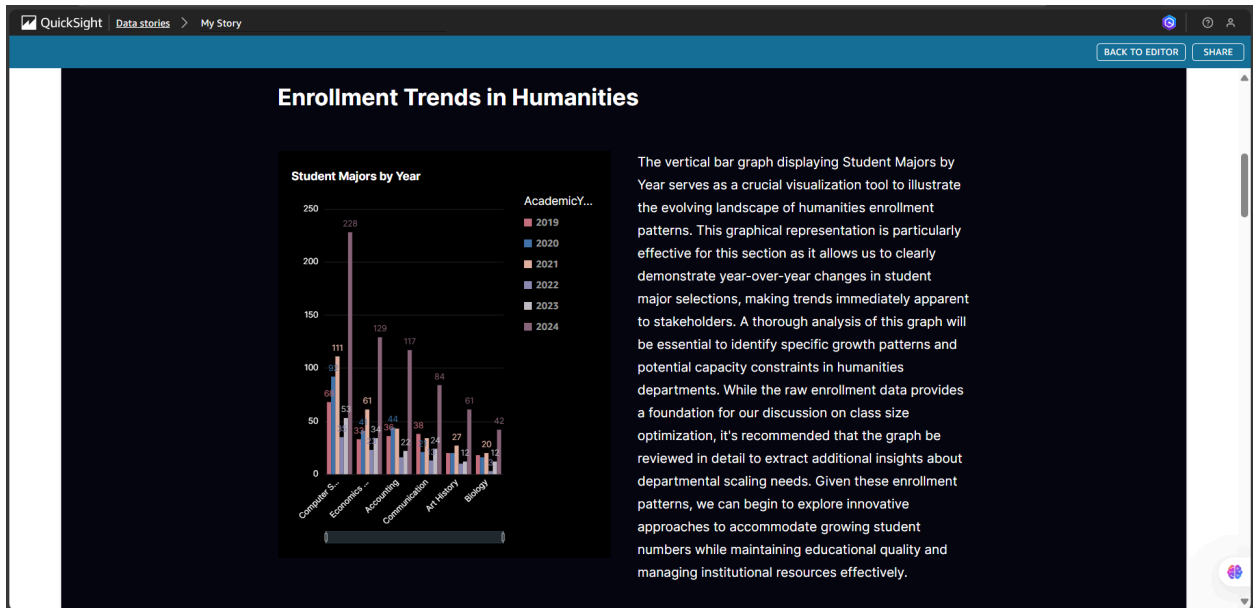
4. Courses with the highest average course costs

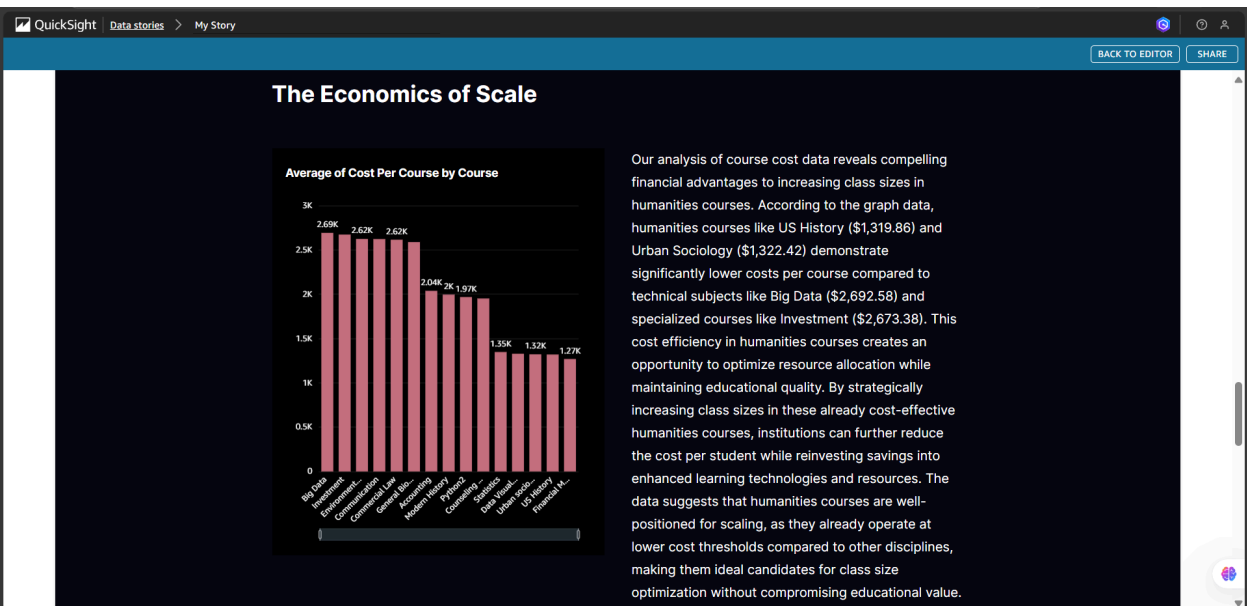


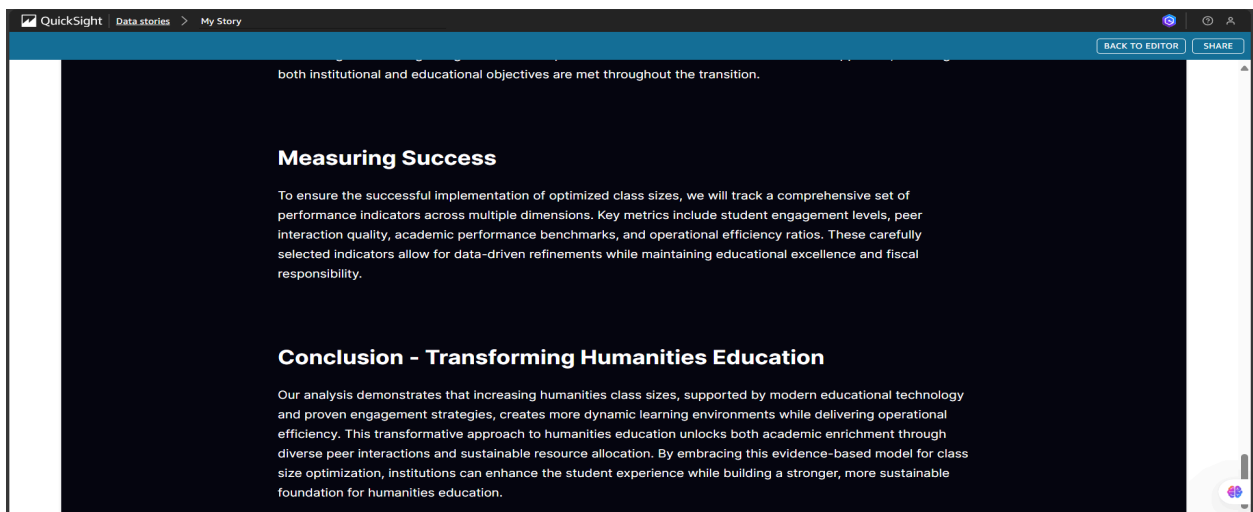
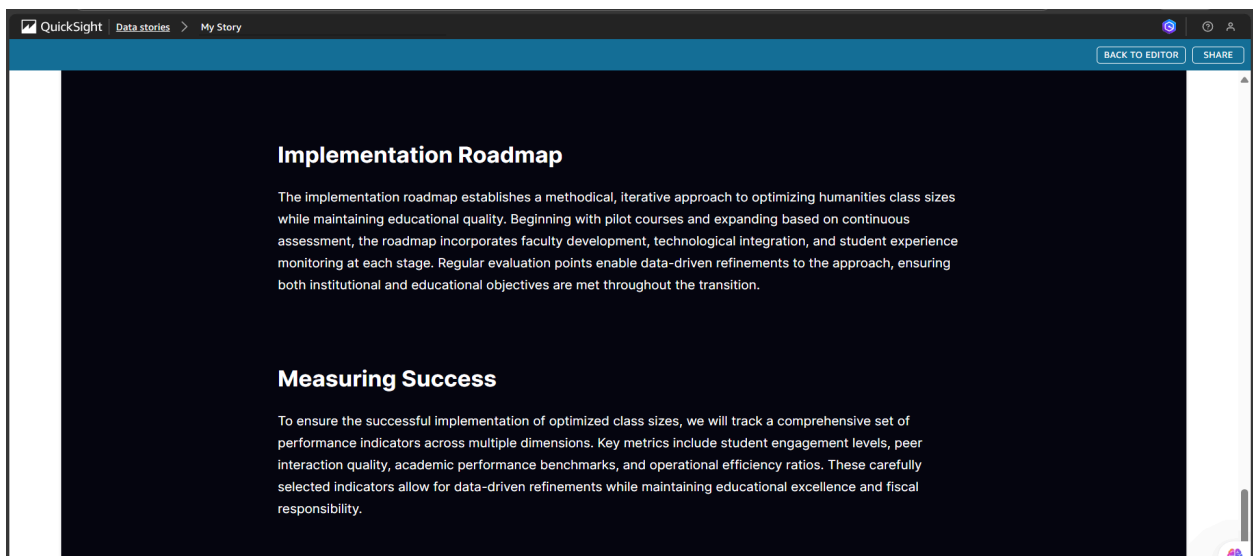
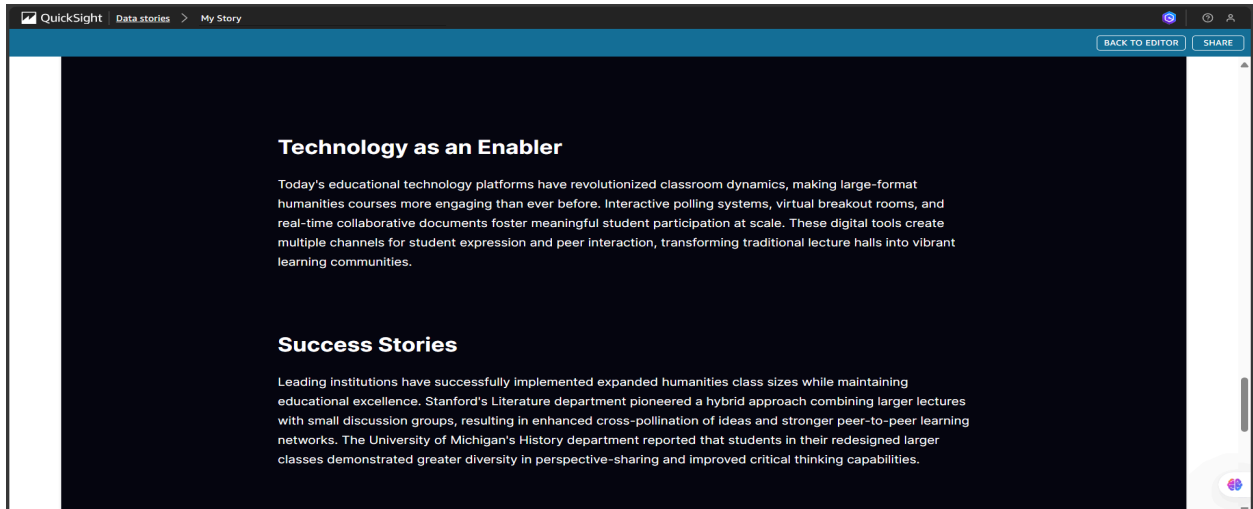
Data Story Creation

A data story was created using Amazon Q to explain the theory behind improving student satisfaction without increasing costs. The story included visuals from the dashboard and was edited to enhance clarity and presentation. Additional text blocks and image blocks were added, and the story style was customized. The cover was updated to include the student's name.









This report includes all required elements: refresh schedule, calculated field formula, visuals, named entities, verified questions, dashboard, scenario thread, and the complete data story. Screenshots have been inserted to support each step.

Conclusion

This project provided a comprehensive walkthrough of Amazon QuickSight and Amazon Q, demonstrating how modern BI tools can transform raw student data into actionable insights for institutional decision-making. From dataset selection to configuring refresh schedules, permissions, and calculated fields, each step reinforced the importance of precision and adaptability in data workflows. The creation of visuals, dashboards, scenarios, and a data story showcased the power of visual analytics and narrative-driven exploration in uncovering trends and guiding strategic recommendations. Despite working with fictional data, the exercise mirrored real-world challenges and solutions, highlighting the value of iterative analysis and thoughtful presentation. Overall, this project not only strengthened technical proficiency in cloud-based BI platforms but also emphasized the role of data storytelling in driving meaningful educational outcomes.